



Effects of perioperative electroacupuncture on postoperative gastrointestinal recovery after thoracoscopic lung surgery

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ABSTRACT

Objective: To study whether perioperative electroacupuncture (EA) can improve postoperative gastrointestinal recovery in patients receiving thoracoscopic lung surgery.

Methods: This study was a single-center, prospective, randomized open-label trial. 180 patients who underwent video-assisted thoracoscopic segmentectomy or lobectomy were randomized to EA group (three sessions, 24 h prior to surgery, postoperative 4 h and 24 h) or usual care group (UC group). The primary outcomes were time to first flatus and defecation. Secondary outcomes included incidence and degree of abdominal distention, postoperative nausea and vomiting (PONV) and pain scores within 72 h after surgery, postoperative morphine use, time to ambulation, and length of hospital stay.

Results: Time to first flatus (15.4 ± 3.2 h vs. 17.0 ± 3.7 h, $P = 0.004$) and time to first defecation (75.9 ± 7.9 vs. 79.7 ± 8.1 h, $P = 0.002$) in the EA group were significantly shorter than the UC group. The incidences of abdominal distention and PONV postoperative 24 h were significantly reduced in the EA group ($P < 0.05$). There was no difference in postoperative pain intensity, morphine use, time to ambulation, and length of hospital stay between the two groups ($P > 0.05$).

Conclusion: Electroacupuncture is a simple intervention for accelerating postoperative gastrointestinal recovery and may be considered as an adjunct strategy in enhanced recovery protocols in thoracoscopic lung surgery.

Video-assisted thoracoscopic surgery (VATS) combined with the Enhanced Recovery after Surgery (ERAS) has been shown to be associated with short hospital stay and a quick and safe return to normal activity¹. Opioid sparing multimodal analgesia, minimally invasive surgery, fewer tubes and intravenous lines are important components of enhanced recovery protocols. Despite these efforts, many patients still suffer transient impairment of gastrointestinal motility after thoracic surgery, which compromises postoperative recovery². Nearly one third of patients reported that their daily life was affected by constipation after discharge, and constipation is often an overlooked aspect of patient care³. In our previous follow-up, gastrointestinal dysfunction after thoracic surgery was common, which was uncomfortable for patients, and affected postoperative quality of life⁴. From this perspective, reducing duration of gastrointestinal dysfunction is regarded as an essential part of the ERAS protocols.

Postoperative ileus (POI), as a clinical manifestation of impairment in gastrointestinal motility following surgery, is characterized by pain,

transient postoperative nausea and vomiting (PONV), inability to pass flatus or stool and abdominal distension^{5–7}. POI is highly inconvenient, embarrassing and uncomfortable for patients, and is closely related to poor quality of life and increased healthcare cost^{6,7}. A retrospective study conducted on participants undergoing thoracic surgery showed that more than 25 % of participants suffered from varied levels of postoperative constipation within 30 days after surgery². Delayed gastrointestinal motility was observed both on the 1st and 3rd postoperative days regardless of the postoperative analgesic technique used². The precise mechanism and cause of POI remains unclear. Factors associated with the development of POI may include inflammation response induced by surgery and anesthesia, long duration of surgery, hemorrhage, activated sympathetic tone, systemic opioid analgesia, mental status and bowel edema^{5,8}. Furthermore, changes in normal diet, less immobilization and unfamiliarity with surroundings can also lead to changes in defecation patterns³. Due to the complex etiological factors of POI, there remains room for improvement in the prevention and

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treatment of POI.

Many strategies have been proposed to improve POI, such as preoperative carbohydrate loading, early postoperative feeding, fluid restriction and minimally invasive surgery, neostigmine, laxatives, erythromycin, naloxone, metoclopramide, chewing gum and prokinetic agents^{9,10}. However, none offer a convincing benefit, nor are they free from side effects. Acupuncture, a well-accepted potential alternative therapy, can offer certain beneficial effects, such as alleviating preoperative anxiety and reducing postoperative pain and postoperative nausea and vomiting (PONV)^{11–13}. Electroacupuncture (EA), which is adding electrical stimulation to the common manual needling, has recently been used as a treatment option for accelerating the recovery of gastrointestinal function after several types of abdominal surgery^{14–16}. Systemic review concluded that EA was a safe and effective treatment for POI after abdominal surgery¹⁷.

Few well randomized trials have assessed the effectiveness of EA on gastrointestinal function in patients undergoing thoracoscopic surgery. This study aimed to evaluate the effect of EA on gastrointestinal recovery in patients who underwent minimally invasive thoracic lung surgery.

Patients and methods

This study was approved and audited by Ethics Committee of Jiangsu Province Hospital of Chinese Medicine, and was registered in the Chinese Clinical Trial Registry (ChiCTR1900024840). A single-center, prospective, randomized study was conducted on patients who underwent thoracoscopic lung surgery from September 1, 2019 to October 20, 2020. Patients were provided by the principal investigator with details regarding the assessment procedure, and gave informed written consent before their scheduled surgery.

Selection criteria

Inclusion criteria were patients scheduled for elective video-assisted thoracoscopic lobectomy or segmental resection surgery, age older than 18 years, American Society of Anesthesiologists grade I–II, and willingness to sign the informed consent. Exclusion criteria were as follows: patients with severe systemic disease, such as cardiovascular, hepatic, or renal abnormalities before operation; patients with cognitive dysfunction, psychological disorder; history of prior esophageal or gastric surgery; history of EA treatment or allergic to acupuncture needles, pregnant or breastfeeding women; use of electrical stimulation devices (pacemaker or implantable defibrillator) severe infection or surgery around the local site of needling, chronic opioid use, conversion to open surgery, failure for paravertebral block (PVB).

Randomization and blinding

Patients were randomly assigned to either usual care(UC) or EA group by computer-generated codes and sequentially numbered, opaque envelopes. Given the characteristic of EA, the acupuncturist and patients were aware of the treatment allocation. Nevertheless, anesthetists, assessors, the data collector as well as statisticians were blinded to treatment allocation throughout the entire study.

Anesthetic proceedings

All surgeries were performed by senior surgeons using minimally invasive techniques by VATS. Anesthetic and surgical techniques were standardized. Anesthesia was induced with propofol 2 mg/kg and sufentanil 0.4 µg/kg. Tracheal intubation was facilitated with 0.1 mg/kg vecuronium. Anesthesia was maintained with target-controlled infusions of propofol and remifentanyl according to the hemodynamics and bispectral index. Ondansetron 4 mg and dexamethasone 5 mg were given before skin closure for prevention of PONV. Intravenous parecoxib

40 mg was used as adjunctive pain management at the end of surgery. The endotracheal tube was removed after adequate spontaneous ventilation and movement resumed.

Under ultrasound guidance, paravertebral nerve block with 20 ml of 0.375 % ropivacaine were used for all patients. Both groups received routine oral celecoxib 200 mg every 12 h and the same postoperative intravenous patient-controlled analgesia (PCA). The PCA consisted of morphine 1 mg/kg, ketamine 1 mg/kg and tropisetron 10 mg, set at a basal rate of 2 ml/h, with a bolus dose of 0.5 ml and a lockout time of 15 min. When patients reported a pain VAS score ≥ 4 or they requested an analgesic, additional bucinnazine 100 mg will be given intramuscularly.

Electroacupuncture(EA) interventions

EA was performed on patients laying supine by our team composed of experienced acupuncturists with at least 10 years of practice. The selection of acupoints were bilateral PC6 (*Neiguan*), ST36 (*Zusanli*), and ST37 (*Shangjuxu*), based on expert consensus provided by qualified and experienced acupuncturists and published literature^{14–18}. Specific acupoints are associated with certain regional effects. The locations of these acupuncture points are described in *The National Standards for Acupoint Location*¹⁹. Patients in the EA group received acupuncture 24 h prior to surgery, 4 h and 24 h after surgery, and each session lasted 30 min. After disinfection, single-use, sterile acupuncture needles (size 0.30×40 mm, Suzhou Medical Supplies Factory Co., Ltd., China) were inserted perpendicular into skin with a depth of approximately 20 to 45 mm. The needles were manipulated via lifting and rotation until the patient felt a *De-qi* sensation, defined as a feeling of heaviness, swelling or numbness around the acupoint²⁰. The stimulator device we used was Huatuo Electronic Acupuncture Treatment Instrument (EA stimulator, XS-998B, Nanjing Xiaosong Medical Instrument Research Institute, Nanjing, China). The ipsilateral PC6, ST36 and ST37 will be separately connected to one electrode set, and therefore yielding three electrode sets. The stimulation frequency was 15 Hz, with intensities adjusted to trigger slight vibration of the needle handles according to individual tolerance.

The EA group also received usual care (UC).

Usual care

Usual care was the routine treatments after thoracic surgery, including routine nursing care, analgesics, dietary instructions and discharge planning²¹. Early ambulation was encouraged. Oral feeding was resumed as early as possible. Participants who experienced absence of flatus over last 72 h postoperatively were instructed to use laxatives. PONV was treated with intravenous ondansetron 4 mg as needed.

No acupuncture treatment was performed in the UC group.

Outcome measures

Primary outcomes of this study were time to first flatus and first defecation.

Times to first flatus and defecation measured in hours were recorded by the patients and then reported to our research assistant.

Secondary outcomes included incidence and degree of abdominal distension (AD), use of the rescue medication, incidence of PON and POV, VAS pain scores, postoperative morphine use, time to ambulation and length of hospital stay. Abdominal distension was measured by a 5-point Likert-type scale ranging from “0 = I feel no abdominal distension at all” to “4 = I feel horrible abdominal distension and need it addressed”^{8,16,21}. Pain intensity was rated on a 0–10 visual analogue scale (VAS, 0 = no pain, 10 = worst pain). Incidence of PON and POV (defined as at least one episode of nausea or vomiting that occurred), pain at rest and on movement were recorded at 0–24 h, 24–48 h and 48–72 h after surgery. Postoperative opioid use for the first 24 h and 72 h after surgery was extracted from the PCA device records.

Statistical analysis

A sample size calculation was performed based on our feasibility study⁴. In that study, the mean reduction of postoperative time to first flatus was 3.3 h, favoring the EA group. To achieve a power of 0.9 and two-sided α of 0.05, the sample size for this study was targeted as 73 in each group. Considering a possible dropout rate of 20 %, a sample size of 184 patients in total was needed in the study.

We conducted our analyses using the intention-to-treat principles regardless of whether patients had received the assigned intervention or completed all EA sessions. Statistical analyses were performed using SPSS22.0 for Windows (SPSS, Chicago, IL, United States). Continuous data were expressed as mean(standard deviation) and median (interquartile range), and categorical variables were expressed as frequency (percentage). *T* tests or Mann–Whitney *U* tests were used to compare the group differences on continuous variables as appropriate. Fisher test and the χ^2 test were used to analyze categorical variables as appropriate. All tests were two-sided, and $P<0.05$ was considered statistically significant.

Results

A total of 184 patients were screened in the study, of whom, 180 were randomly assigned to EA group ($n = 90$) or UC group ($n = 90$). After randomization, 3 patients withdrew from the trial because they were declined to continue EA on postoperative day 2 ($n = 1$), didn't receive PVB ($n = 1$) or converted to open surgery ($n = 1$). These three patients were retained in the final analysis. Finally, 90 patients per group were included in the intention-to-treat analyses (Fig.1). The baseline demographic data and clinical characteristics of the two groups were comparable and did not differ significantly (Table1).

The time to first flatus was 2 h shorter in the EA group as compared with that of the UC group ($P = 0.004$). The time to first defecation was more than 4 h shorter, than UC group, $P = 0.002$ (Table 2).

The incidence of abdominal distension was significantly reduced in

Table 1
Demographic data and surgical details.

Variables	EA group	UC group	<i>P</i>
Age(yr) ^a	52.7 ± 11.0	55.8 ± 11.6	0.07
Weight(kg) ^a	62.8 ± 10.0	61.8 ± 10.0	0.50
Sex (n[%]) ^b			0.11
Male	24(26.7)	34(37.8)	
Female	66(73.3)	56(62.2)	
Patients with comorbidities	21(23.3)	24(26.7)	0.61
Surgical procedures			0.09
Lobectomy (n[%]) ^b	29(32.2)	40(44.4)	
Segmentectomy (n[%]) ^b	61(67.8)	50(55.6)	
Duration of operation (min) ^c	90(68.8–126.3)	90(70–140)	0.61
One lung ventilation time(min) ^c	60(40.0–95.0)	59(40–100)	0.90
I.V. fluid(ml) ^c	1000(800–1000)	1000(897–1000)	0.16
Blood loss(ml) ^c	50 (20–80)	50(20–100)	0.30
Intraoperative opioids			
Sufentanil(ug) ^c	32.5(30–35)	32(30–35)	0.79
Remifentanyl(mg) ^c	0.5(0.3–0.6)	0.5(0.4–0.7)	0.75
Duration of chest tube(d) ^c	3(3–4)	3(3–5)	0.39

EA: electroacupuncture; UC: usual care; I.V.: intravenous.
Data were expressed as mean±SD^a, n(%)^b or median (quartile range)^c.

the EA group (34.4% vs. 51.1 %, $P = 0.024$). There was a trend toward decreased degree of abdominal distension, but did not reach a significant difference ($P = 0.069$). Patients who required rescue medication were also significantly less in the EA group than UC group (53.3% vs. 71.1 %, $P = 0.014$) (Table 2).

26.7 % participants in the EA group, compared with 44.4 % in the UC group experienced PON in the 0–24 h after surgery. Also, the incidence of 0–24 h POV(13.3 %) in Group EA was significantly lower than that in Group UC(27.8 %). There were no significant differences in the incidence of PONV at 24–48 h.

Post operative pain was minimal, about 2/10 in both groups. There were no significant differences in the post-operative pain at rest or on movement between groups. Cumulative morphine consumption during the postoperative day 1 and 3 were similar between the two groups.

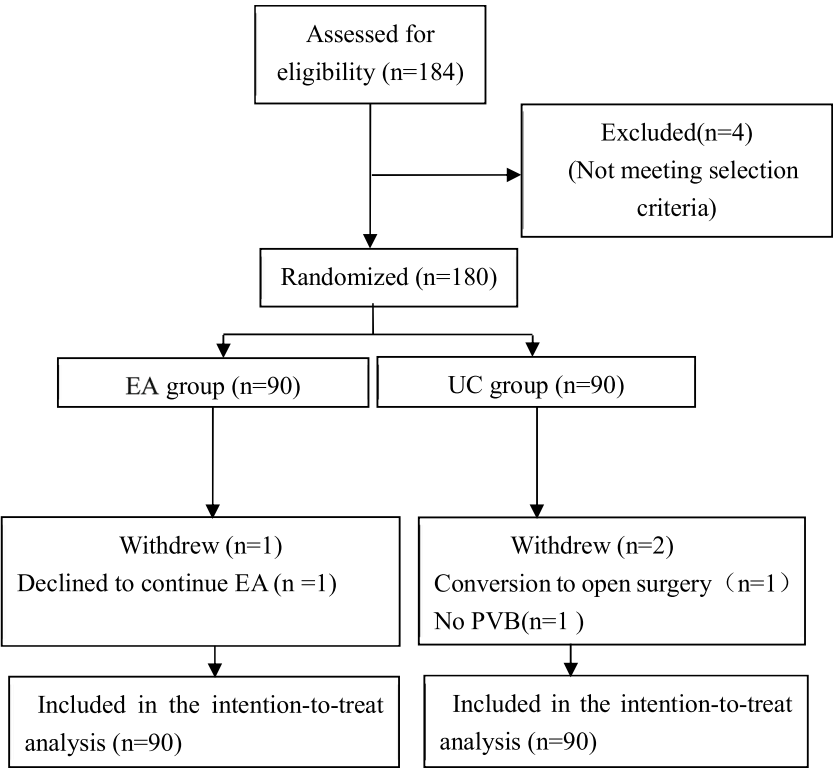


Fig. 1. Flowchart of participants through the study.

Table 2
Outcomes on gastrointestinal function.

Variables	EA group	UC group	P
Time to first flatus (h) ^a	15.4 ± 3.2	17.0 ± 3.7	0.004
Time to first defecation (h) ^a	75.9 ± 7.9	79.7 ± 8.1	0.002
Rescue medication (n [%]) ^b	48(53.3)	64(71.1)	0.014
AD's incidence (n [%]) ^b	31(34.4)	46(51.1)	0.024
AD's degree			0.069
0	59(65.6)	41(45.6)	
1	16(17.8)	42(46.7)	
2	13(14.4)	4(4.4)	
3	2(2.2)	3(3.3)	
4	0(0)	0(0)	

EA: electroacupuncture; UC: usual care; AD: abdominal distension.
Data were presented as mean ± SD^a or n[%]^b.

Time to ambulation and length of hospital stay did not differ between two groups. (Table 3).

Discussion

This trial examined the effect of perioperative EA on gastrointestinal function after thoracoscopic lung surgery. Our results showed that perioperative EA led to a significantly shorter time to first flatus and first defecation, decreased incidence and a trend toward alleviated abdominal distension and reduced incidence of PONV during first 24 h, reflecting a multi-facet effect of EA on the gastrointestinal functional recovery.

Constipation and impaired defecation pattern following thoracic surgery contributes to delayed recovery^{3,4} and costs to the healthcare system²², and is often overlooked in patient care³. Systemic prokinetic drugs are often used for POI, but the evidence is insufficient to recommend the use of cholecystokinin-like drugs, cisapride, dopamine antagonists, propranolol or vasopressin²³. Effective non-pharmacological effort that can reduce patient suffering and improve recovery should be properly tested and encouraged. Our study showed that EA, in addition to UC, was associated with shorter time to first flatus and first defecation. Also, EA group showed decreased incidence and degree of abdominal distension and rescue medication. This result was in consistent with previous randomized trials and meta-analyses demonstrating positive effects of acupuncture in abdominal surgery^{14–17}. Our results suggested that EA offered additional clinical benefit to patients undergoing thoracoscopic lung surgery, but placebo effects cannot be excluded without a sham acupuncture control²⁴.

Table 3
Other clinical outcomes.

Variables	EA group	UC group	P
0–24 h after surgery			
Incidence of Nausea [n(%)] ^b	24(26.7)	40(44.4)	0.013
Incidence of Vomiting[n(%)] ^b	12(13.3)	25(27.8)	0.016
Pain VAS at rest ^a	2(2–3)	2(2–3)	0.844
Pain VAS at movement ^a	3(3–4)	3(2–3)	0.159
24–48 h after surgery			
Incidence of Nausea [n(%)] ^b	6(6.7)	14 (15.6)	0.058
Incidence of Vomiting[n(%)] ^b	6(6.7)	10 (11.1)	0.295
Pain VAS at rest ^a	2(2–2.5)	2(2–2)	0.390
Pain VAS at movement ^a	3(2–3)	3(2–3)	0.652
48–72 h after surgery			
Incidence of Nausea [n(%)] ^b	0(0)	0(0)	–
Incidence of Vomiting[n(%)] ^b	0(0)	0(0)	–
Pain VAS at rest ^a	2(2–3)	2(2–2)	0.353
Pain VAS at movement ^a	3(2–3)	3(2–3)	0.314
Morphine use on POD 1(mg) ^c	43.7 ± 4.7	44.0 ± 4.9	0.733
Morphine use on POD 1–3(mg) ^c	127.0 ± 12.0	127.5 ± 12.5	0.785
Time to ambulation (h)	6(5–7)	6(5–7)	0.208
Length of hospital stay (d)	6(5–7)	6(5–7)	0.202

EA: electroacupuncture; UC: usual care; VAS: Visual Analogue Scale.
Data were expressed as median (quartile range)^a, n(%)^b or mean±SD^c.

The selection of acupoints for our study were PC6, ST36, and ST37, which was based on evidence from published studies and the theories of Chinese medicine^{19,25,26}. EA at PC6 not only has antiemetic effects, but also increases gastric motility via promoting efferent vagus nerve activity^{27,28}. ST36 and ST37 are located along the “Stomach Meridian” of Foot-Yangming on the anterior-tibia region. ST36 is at the junction of the Stomach Meridian and plays an important role in balancing Yin and Yang in the abdomen to relieve distention and constipation. Stimulation at ST36 can enhance the small intestinal motility, possibly by down-regulating the concentration of vasoactive intestinal peptide and up-regulating the concentrations of motilin, ghrelin and gastrin²⁹. ST37 is an acupoint that mainly regulates colon motility to assist gas passage. EA stimulation at ST37 increases defaecation frequency and gastrointestinal transit markers of constipation, such as serotonin and tryptophan hydroxylase concentrations²⁶. In general, EA improved the gastrointestinal tract transit function by activating the vagal nerve directly and regulating peripheral inflammation^{27,28,30}. A combination of these three points are the preferred acupoints for treating POI, contributing to the encouraging results of our study.

The optimal timing and duration of EA in thoracic surgery has not been studied before. The majority of studies used EA postoperatively until the first flatus or defecation^{14–17,31}. In one study conducted by Li S et al., one-session EA 24 h prior to the surgery also accelerated motility of the gastrointestinal track³². Liu MY et al.²¹ found that, when compared with no acupuncture treatment, EA 24 h before surgery reduced the incidence and degree of abdominal distension and shortened the time to first defecation and duration of hospitalization after vascular laparotomy. In our study, EA stimulation was applied both pre- and post-operatively. Consistent with our feasibility study⁴, most patients were satisfied with the frequency and duration of acupuncture, and only one patient discontinued EA postoperatively due to discomfort. Our protocol of three treatment sessions may seem to demand patients' time, it was well accepted⁴. From clinical point of view, it might be easier to use single session before or after surgery. Future research is needed to identify optimal timing and regimen of EA to enhance gastrointestinal recovery, to assess potential patients' satisfaction and cost-effectiveness with this intervention.

Studies have used different endpoints to record POI. It is important to define sensitive and objective measures that accurately captures the spectrum of POI. Time to first flatus and defecation are easy and simple components in the assessment of gastrointestinal dysfunction and POI^{21,25,31,33}. However, available evidence for these indicators is limited. For example, time to first flatus is difficult to assess accurately and time to first defecation might simply reflect rectal emptying and provide no reliable information on the recovery of whole gut. Bowel movements are the most reliable endpoint, real-time bowel sound recognition using wearable health monitoring should be used in future trials. In our institution, early oral feeding as part of ERAS is started without the return of bowel sounds after thoracic surgery, so we didn't monitor intestinal sounds.

PONV is common in surgical patients and represents a frequent and critical cause of delayed recovery. Acupuncture on P6 has been widely used to reduce PONV after several types of surgery^{11,13}. Despite PONV prophylaxis with dexamethasone and ondansetron, PONV incidence was still high in both groups. The 44.4 % and 27.8 % incidence of respective PON and POV in the UC group were a bit higher, compared with 16.9 % reported by the literature on this form of surgery³⁴. Individual susceptibility and opioid use such as morphine PCA are strongly regarded as the most likely causes of PONV in this study. It is pleasing to see in our study, EA significantly reduced the incidence of nausea and vomiting on the 1st postoperative day. Combination with EA and routine antiemetics may offer additional benefit for those patients.

Opioid use was associated with PONV and may reduce the time required to restore normal gastrointestinal function³⁵. Sun Y, et al. found that acupuncture treatment reduced opioid consumption, pain intensity and opioid-related side-effects³⁶. On the contrary, we didn't

find any beneficial effect of EA on pain scores and opioid consumption after surgery. A few reasons might explain this: 1) our study only focus on thoracoscopic lung surgery, which is a less painful surgery, when compared with open abdominal surgery. In Sun's review, all forms of surgeries were included; 2) we used multimodal analgesia to relieve pain and minimize opioid use. Multimodal analgesia was achieved with nonsteroidal anti-inflammatory drugs, PVB and morphine PCA, producing sufficient analgesia after VATS with pain score less than 4/10³⁷. In addition, the use of PVB might have underestimated the analgesic effects of EA by blocking both somatic and sympathetic neural pathways, which might be considered as a confounding factor on the effects of EA³⁸.

Postoperative gastrointestinal recovery is an crucial part in the enhanced recovery of surgical patients. Ng SS, et al. demonstrated that EA was associated with shorter hospital stay after surgery, when a fast-track perioperative program was not available in their hospital¹⁶. Liu T, et al.³⁹ reported that ambulation within 2 h of extubation was safe and feasible and could lead to shorter length of hospital stay and time to first flatus in patients with undergoing thoracoscopic lung surgery. In our study, both groups had similar time to ambulation and hospital stay, because we applied enhanced recovery program during our routine clinical practice.

There were certain limitations to this study. First, a placebo effect can't be excluded without a sham EA group. Because sham EA was not used in this study, the inadvertent interaction between assessors and patient in the UC group may affect the results. However, according to a Cochrane review, inclusion of a sham EA group would not have changed the research conclusions⁴⁰. Second, lack of standardized outcome measures may limit the interpretation of the findings. Finally, this study only investigated the effect of EA on bowel motility, PONV and postoperative pain in a short period. The incidence of postoperative constipation and chronic pain may last for a long time after surgery^{3,41}, so the long term effects of EA on gastrointestinal function and chronic pain in patients undergoing thoracic surgery needs further study.

Conclusion

Despite certain limitations, we provide evidence that perioperative EA, in combination with standard usual care, has multi-facet effect and can lead to a faster recovery of gastrointestinal function following thoracoscopic surgery.

Author contributions statement

Jie Yang, Fangbing Ji and Man Zheng: Study concept and design. **Jie Yang, Libing Huang, Juan Zhu and Siying Liu:** Acquisition of data. **Jie Yang, Weiqian Tian and Libing Huang:** Analysis and interpretation of data. **Zhen Zheng:** Study design, acupuncture design, manuscript revision. **Man Zheng:** Supervisor of the study

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Ethics approval

This study was approved and monitored by the Institutional Review Board of Jiangsu province hospital of Chinese Medicine (2017NL-123-02).

Consent to participate

Informed consent was obtained from all individual participants included in the study to participate.

Consent to publish

Informed consent was obtained from all individual participants included in the study for findings to be published.

Availability of data and material

Will be made available upon request to the corresponding author.

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